

Inside a Firing Neuron: Ion Channels and Action Potentials

PhET Neuron · PhET

Senior Secondary (Years 11-12)

~30 minutes

Psychology

NOTICE — AI-generated worksheet

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Simulation: <https://phet.colorado.edu/en/simulations/neuron>

Curriculum links: brain function · sensation · memory

Learning intentions

- I can describe the sequence of ion channel activity that produces an action potential.
- I can explain how sodium and potassium movement changes the neuron's membrane voltage.
- I can relate the biological process of neural firing to psychological processes such as sensation.

Getting started

Open the Neuron simulation and locate the stimulus control near the axon. Stimulate the neuron once and watch the full sequence play out without pausing, then stimulate it again and use the pause or step controls to work through the stages slowly for the tasks below.

Tasks

1. Stimulate the neuron and describe, in your own words, what happens to the membrane voltage graph.

2. Pause the simulation at the point where the voltage rises sharply. Which ion channels are open at this stage, and which ion is flowing?

3. Step forward to the point where the voltage falls back down. Which ion channels are now doing the work, and which ion is flowing?

4. Explain, in sequence, how the opening and closing of sodium and potassium channels together produce the rise-and-fall shape of the action potential graph.

5. After an action potential fires, there is a brief period where the neuron cannot fire again (the refractory period). Use the simulation to explain what is happening with the ion channels during this stage.

6. Try stimulating the neuron with a very weak stimulus. Does it produce a full action potential? What does this suggest about how neurons respond to weak signals?

7. This 'all-or-nothing' firing pattern is a basic unit of how the nervous system processes information. Explain how a stronger sensory stimulus (for example, a brighter light or louder sound) could still be represented by neurons, given that each individual action potential is the same size.

Extension (optional)

Reason from what you know of neuron structure: how does a signal travel from one neuron to the next, and why can't the electrical process you observed in this simulation cross that gap directly?
